Governing by Artifacts: An Auditable Constraint-Based Framework for Contestable Educational Decision Systems

Tales Matos
School of Engineering
University of Guelph
50 Stone Rd E
Guelph, ON N1G 2W1, Canada
tbenigno@uoguelph.ca

Victor Oliveira Santos
School of Engineering
University of Guelph
50 Stone Rd E
Guelph, ON N1G 2W1, Canada
volive04@uoguelph.ca

Bahram Gharabaghi*
School of Engineering
University of Guelph
50 Stone Rd E
Guelph, ON N1G 2W1, Canada
bgharaba@uoguelph.ca

Jesse Van Griensven Thé
Lakes Environmental
170 Columbia St. W
Waterloo, ON N2L 3L3, Canada
Jesse.the@weblakes.com

Jose Antonio Macedo
Departamento de Computacao
Federal University of Ceara
Fortaleza, CE 60020-181, Brazil
jose.macedo@insightlab.ufc.br

ABSTRACT

Educational decision systems need inspectable evidence linking decisions to declared policies. We present Compass-ECL-MAS, an artifact-oriented framework combining empirical constraints, deterministic selection over feasible Pareto candidates, versioned policies, and replay of Top-K decisions with abstention. Its export contract requires both verified evidence and satisfaction of declared empirical restrictions. We evaluate this contract using fixed correctness or student-success scores from ASSISTments, OULAD and a synthetic stress test. Policy sweeps show how capacity, uncertainty thresholds and the treatment of unselected cases affect feasibility and automation coverage. In a retrospective selection/test partition, ASSISTments and OULAD satisfy the sample constraints; synthetic validation has no feasible policy. Seven synthetic fault fixtures show that replay agreement can coexist with incomplete evidence, and verified evidence can coexist with a constraint violation. An exploratory counterbalanced study ($N = 16$; eight standardized cases) finds diagnostic and remediation accuracy of 0.5625 with logs/cards and 0.921875 with structured evidence; the two correctness endpoints coincide in every response. The study evaluates the evidence bundle as a whole. The framework provides contestability-enabling infrastructure through reproducible policy analysis, evidence checks and explicit export rejection. Its demonstrated scope is fixed-score decision reconstruction, with institutional appeal processes and prospective educational effects requiring separate evaluation.

CCS CONCEPTS

• **Computing methodologies** → **Artificial intelligence**; • **Social and professional topics** → **Computing / technology policy**; *Government technology policy*.

KEYWORDS

contestability, algorithmic accountability, procedural justice, public sector AI, educational decision support, fairness, audit trails, policy versioning, abstention

1 NOTATION AND CORE DEFINITIONS

Table 1 summarizes the symbols used throughout the paper. We emphasize that all fairness metrics are defined *under the specified decision rule* (Top-K with abstention), not merely on raw scores.

2 INTRODUCTION

Educational decision systems support the allocation of tutoring, counseling, assessments and other scarce resources. Their institutional consequences extend beyond predictive accuracy: students and families need to understand the rule applied to a decision, identify errors and present evidence for reconsideration. An audit therefore needs to connect a case to the policy, inputs and constraints that produced its outcome.

Public controversies illustrate the consequences of weak procedural safeguards. The UK A-level grading algorithm in 2020 generated socially salient downgrades and a policy reversal [27]. Wisconsin's dropout early-warning system drew scrutiny over inequitable risk assignments and opaque governance [9]. In adjacent public-sector risk profiling, the Dutch SyRI ruling highlighted

*Corresponding author.

Table 1: Notation summary.

Symbol	Meaning
n	number of instances in a dataset split
x_i, y_i, a_i	features, label, sensitive attribute for instance i
f_θ	predictive model with parameters θ
s_i	score $s_i = f_\theta(x_i) \in [0, 1]$
u_i	predictive entropy in bits, $0 \le u_i \le 1$
γ	abstention threshold (eligibility)
$\mathcal{E}_\gamma$	eligible set $\{i : u_i \le \gamma\}$
K	selection capacity ceiling (Top-K)
TopK$(\cdot)$	selected set under Top-K
C	automation coverage $(n_{\text{sel}} + n_{\text{rej}})/n$
$\Delta_{\text{EOpp@K}}$	equality of opportunity gap under Top-K

transparency and accountability requirements [7]. These cases motivate decision evidence that institutions and affected people can inspect and challenge.

Model-level metrics provide only part of that evidence. Selective indicators can support claims of ethical compliance while leaving responsibility for trade-offs and access to recourse unclear [5, 15, 20, 26]. Educational labels and measurements can also encode structural inequities [3]. Governance consequently requires explicit procedures for choosing policies, checking decisions and handling failures.

We present Compass-ECL-MAS, a layer for translating declared requirements into versioned decision policies and auditable artifacts. ECL checks empirical restrictions; MAS selects a policy from feasible Pareto candidates using ordered criteria and deterministic tie rules. Per-case packets preserve the evidence required to reconstruct the supported decision rule. A verifier checks that evidence, and a technical gate permits export only when verification and the declared empirical constraints pass. Abstention produces a record requiring review; connecting that record to an institutional appeal process remains a deployment responsibility.

2.1 Contributions

The paper makes three connected contributions:

- **An executable export contract.** We specify how fixed-score replay, evidence verification and empirical feasibility jointly determine technical export permission. Nested acceptance predicates make the conditions and failure reasons explicit.
- **Reproducible policy and failure analyses.** Fixed-score sweeps and a retrospective selection/test partition examine policy-dependent feasibility. Seven synthetic fault fixtures test when decision agreement is insufficient for evidence acceptance or export permission.
- **Exploratory evidence on structured contestation materials.** A counterbalanced study compares a complete evidence bundle with logs/cards on standardized diagnosis and remediation tasks, with paired estimates and explicit methodological limits.

2.2 Scope

The contribution concerns governance infrastructure: explicit rules, deterministic policy selection, preserved evidence and observable rejection of exports that fail the contract. The framework combines established selection, provenance and integrity mechanisms; the experiments evaluate their specified acceptance conditions in a reference application. Human task performance and computational integrity are evaluated separately.

The predictors estimate interaction correctness or student success. They provide testbeds for policy mechanics, rather than validated rules for allocating support to the students at greatest risk. The human study evaluates the structured-evidence bundle without isolating its individual components. Accessible appeal channels, legitimate policy adoption, staffing and remedies remain institutional requirements. The demonstrated scope is therefore contestability-enabling infrastructure; prospective early-warning targeting and educational treatment effects require separate evaluation.

3 RELATED WORK

FAccT research has repeatedly shown that "getting the metric right" is insufficient when systems are embedded in institutions with asymmetric power, weak recourse, and opaque accountability. We position Compass-ECL-MASat the intersection of (i) contestability and procedural justice for algorithmic systems, (ii) principle-to-practice governance, and (iii) model-level fairness tooling and trade-offs.

3.1 Contestability, Due Process, and Institutional Accountability

A central FAccT concern is whether affected people can *understand, challenge, and obtain remedy* when algorithmic systems shape life chances. Citron and Pasquale argue that scoring systems can erode due process by obscuring decision logic and shifting power toward institutions [5]. Subsequent work emphasizes that accountability is not a property of an algorithm alone, but of the socio-technical system: what is documented, what is disclosed, who can audit, and what remedies exist [15, 20, 26]. In HCI and socio-technical design, contestability-by-design frameworks articulate practices that keep systems responsive to human intervention throughout the lifecycle [1].

Our contribution aligns with this line: we treat *audit artifacts* (policy manifests, constraint reports, decision traces) as first-class outputs required for contestation and institutional oversight.

3.2 From Principle-Level Ethics to Operational Governance

Major AI governance frameworks—NIST AI RMF, OECD AI Principles, UNESCO, and EU Trustworthy AI guidelines—converge on high-level desiderata (fairness, transparency, accountability, robustness) but remain implementation-agnostic [8, 17, 21, 22]. Scholars have highlighted the "last-mile" gap between ethical principles and operational mechanisms [6, 16]. We build on this insight, but focus on a specific public-sector domain where legitimacy and recourse are central: educational decision support.

Table 2: Claims, supporting evidence and scope of the evaluation.

Claim	Evidence	Boundary
Fixed-score replay	State reconciliation and taxonomy replay, Tables 8 and 12	Does not support changed features or medical exceptions.
Empirical selection	Algorithm 1, Tables 9 and 10	Historical same-sample exploration plus retrospective split (Table 15); no population fairness guarantee.
Explicit failure	Export gate, roster verification and seven fault contracts	Historical violation logging was not operational fail-closed; no human queue connected.
Structured evidence benefit	Tables 5 and 6	Exploratory N=16, no formal ethics review; bundle effect, not isolated replay effect.
Entropy diagnostics	Table 13 and structured group-risk report	Scalar proxy; conditional row intervals, not cluster-robust field validation.

Table 3: Roles of related approaches; this conceptual positioning is not an exhaustive capability audit of all implementations.

Approach	Main role	Relationship to this work
Ethical principles	State institutional commitments	Motivate requirements, without fixing our schema.
Fairness toolkits	Measure or mitigate group disparities	Complement empirical ECL checks.
Impact assessments	Document impacts and responsibilities	Complement policy rationale and review procedures.
Contestability by design	Keep systems open to intervention	Motivates preserved case evidence and bounded replay.
COMPASS–ECL–MAS	Fixed-score policy audit and artifacts	Demonstrates explicit rules, reconstructed states and export gating.

3.3 Fairness Tooling, Trade-offs, and Preference-Based Selection

Model-level toolkits such as Fairlearn and Aequitas provide practical metrics, mitigation strategies, and reporting to support fairness evaluation [19, 25]. These metric-oriented roles are complementary to our decision-evidence contract; we have not exhaustively audited every extension of these toolkits. Trade-offs between utility, fairness, and other constraints are unavoidable; preference-based selection over Pareto-feasible candidates has been explored in multiple forms, including "fairness without harm" style decision rules and policy-oriented selection [23]. Our system-level contribution is to embed such choices into *versioned policy* and to couple selection with a procedural safeguard (abstention to human review) and an artifact trail.

3.4 Algorithmic Systems in Education

Educational prediction systems (early warning, risk scoring, intervention targeting) are widespread, but educational data often contains measurement and historical biases that complicate fairness claims [3]. Our setting therefore emphasizes governance practices (documentation, auditability, and recourse) in addition to model metrics.

Abstention, selective prediction, and distribution-free uncertainty. Our triage setting uses abstention ("reject option") to mark uncertain cases as requiring human review. This idea has a long history in statistical decision theory (e.g., Chow's rule) and has been revisited in modern *selective classification* and distribution-free uncertainty quantification (e.g., SelectiveNet; conformal prediction) [2, 4, 11]. These lines of work motivate two design choices in our system: (i) treating high abstention as a *governance signal* rather than a silent knob, and (ii) recording the abstention policy in a replayable artifact so that institutions can audit when and why automation requested human review.

3.5 Provenance, reconstruction and integrity mechanisms

Provenance and version tracking are established capabilities. PROV-O models entities, activities, agents and derivation relations [14]. ModelDB records models, pipelines, parameters and metrics [24]; Ormenisan et al. use change capture and data versions for pipeline reproducibility and debugging [18]. Our score-only replay has a narrower scope than reconstructing inference from versioned features.

Cryptographic log guarantees also predate this work. RFC 9162 specifies signed transparency-log structures and inclusion/consistency proofs for certificates [13]. Our unsigned batch roster and byte digests do not supply those guarantees. In relation to sociotechnical contestability [1], the present engine supports scrutiny but does not enact an institutional appeal. Appendix 10 compares these specific mechanisms without treating unexamined implementations as lacking features. The contribution is a bounded, executable decision-policy contract and its evaluated distinctions, not a new provenance standard, log-security primitive or proof of first integration.

The contract in Section 4.4 makes the acceptance dependencies explicit and testable. It does not introduce a new provenance primitive: its evaluated claim is that reproducing available decisions can pass while complete evidence verification fails, and that verified evidence can still fail a declared export constraint. Table 17 situates the constituent mechanisms; Table 16 tests these separations in our implementation.

Table 4: Candidate decision states. A separate run-level gate determines whether export is authorized.

State	Accounting
`select`	Within capacity; counts toward selection and automation.
`reject`	Automated non-selection; counts toward automation.
`abstain`	Request for review; excluded from automation coverage.

4 METHOD: POLICY, DECISIONS AND ARTIFACTS

4.1 Design scope

Our design makes institutional choices explicit, preserves evidence for audit, separates scoring from sensitive-attribute auditing, and treats abstention as a request for human consideration. The six COMPASS commitments—dignity, correctness, transparency, harmony, practical wisdom and active compassion—organize requirements; they are a domain-specific design vocabulary, not an exhaustive ethical theory. In the evaluated release the enforced numerical constraints are a fairness-gap cap and an automation-coverage floor. Documentation, intelligibility and institutional recourse remain separate requirements.

4.2 Decision rule and units

A score s_i estimates $\Pr(y_i = 1 \mid x_i)$. All reported rankings are descending. Binary predictive entropy is

$$u_i = -q_i \log_2 q_i - (1 - q_i) \log_2 (1 - q_i), \tag{1}$$

where q_i clips s_i to $[10^{-7}, 1 - 10^{-7}]$ in float64. Entropy and γ are in bits; the maximum is one. Thus $\gamma = 0.999$ need not admit every case. Invalid probabilities and incompatible units are rejected.

The primary rule applies global abstention first: $u_i > \gamma$ yields `abstain`. Among eligible cases, the highest scores are selected up to capacity K. Equal scores are resolved by ascending persistent case ID. K is a ceiling: fewer than K cases may be selected. In allocation mode the remainder is `reject`; in screening mode it is `abstain`. The alternative Top-K-only abstention mode first nominates the global Top-K, abstains uncertain nominees and does not refill their places. These modes are explicit policy fields in the reference implementation.

Metrics. Let n_s, n_r, n_a count select, reject and abstain, with $n_s + n_r + n_a = n$. Automation coverage is $C = (n_s + n_r)/n = 1 - n_a/n$; selection rate is n_s/n. In screening mode $C \le K/n$, so a floor $C_{\min} > K/n$ is structurally infeasible. Changing a deferral into a rejection changes institutional responsibility; it does not demonstrate greater educational benefit.

The *selected-set positive outcome rate* (PosRate; legacy operational name Utility@K and JSON key `utility_at_k`) is $\sum_{i:d_i=\text{select}} y_i/n_s$, undefined for empty selection. Its denominator is the realized selection count, not capacity K. It measures observed correctness or success among selected cases; it is not educational benefit, intervention utility or a causal treatment effect. For binary audit groups we report the absolute gap between $\Pr(d = \text{select} \mid y = 1, a = 0)$ and $\Pr(d = \text{select} \mid y = 1, a = 1)$. Positive-label denominators include all cases in each group, not only eligible cases. A missing group or zero positive denominator produces insufficient evidence, not a zero gap. Labels support retrospective evaluation; they are not inputs required to replay a future decision.

4.3 ECL and the authorization boundary

ECL checks a candidate against declared empirical constraints. The paper sweep uses gap ≤ 0.05 and coverage ≥ 0.50. These are example normative choices, not universal safety thresholds. The historical synthetic recipe used a 0.60 coverage floor; the common replay analysis explicitly uses 0.50. Both reject its initial 0.004 coverage.

The schema 2.0 runner writes a technical export gate: an infeasible or insufficient-evidence result cannot produce an authorized-decision export. Candidate packets remain available for diagnosis. A review-required record is generated, but no human queue is connected and no delivered review is claimed. Here authorization means only technical permission to export under the declared checks. The gate does not verify an institutional mandate, legitimate policy adoption or resolution of an appeal. Statistical feasibility on an audit sample does not certify future population fairness or legal compliance.

4.4 Verifiable export contract

We specify the evaluated export path as nested acceptance predicates. The domain is a supported, nonempty fixed-score batch, an accessible declared policy and its local run metadata. Scores must be finite probabilities, case IDs persistent, and the configured audit metrics computable from the supplied retrospective labels and groups. These assumptions do not establish that source data or policy authors are trustworthy.

Replay and evidence. Let R mean that applying the available policy to the supplied scores and IDs reproduces the recorded decision states. It concerns the supplied cases, not whether the entire expected batch is present. Let V mean that the additional evidence checks pass: required schema, units and clipping convention; policy, manifest, model-byte and packet digests; a unique case roster equal to the expected roster in metadata; and agreement between recorded and recomputed entropy within absolute tolerance 10^{-12}, with zero relative tolerance. The complete verifier accepts exactly when $I = R \wedge V$ on this supported path. In the API, its legacy field `replayable` denotes this complete acceptance, whereas `replay_only` in the fault experiment evaluates only R. A model digest checks bytes without executing the model or proving that it generated the scores.

Constraints and export. Let F mean that the emitted states satisfy the empirical restrictions actually enabled in the declared policy. For the evaluated profiles these are the absolute EOpp-gap cap and coverage floor. A required gap that cannot be estimated yields insufficient evidence and fails this check. Disabling a restriction removes it from F; it does not establish that the omitted requirement holds. The runner grants technical permission exactly when $A = I \wedge F$, recording the gate and exporting authorized decisions only on acceptance. A failed check preserves candidate evidence and records review required, without delivering a human review.

Algorithm 1 Empirical ECL–Pareto–MAS selection

Require: Candidates C, constraints, criteria T, tolerances δ_t
$F \leftarrow$ candidates with finite objectives satisfying constraints
if $F = \varnothing$ **then**
  return infeasible; close authorization gate
end if
$F \leftarrow$ nondominated candidates in F
for criterion t in T **do**
  $v^* \leftarrow$ best value of t in F
  $F \leftarrow \{c \in F : |t(c) - v^*| \le \delta_t\}$
end for
return candidate with smallest persistent ID in F

Separation and limits. Thus $A \Rightarrow I \Rightarrow R$; the predicates are nested, not logically independent. Table 16 supplies counterexamples to both converses: an omitted case preserves R while violating I, and a constraint-violating policy preserves I while violating A. These equivalences are necessary and sufficient conditions for this implemented acceptance path, not for legitimate deployment. Local unsigned metadata anchor completeness; coordinated rewriting can defeat authenticity claims. No predicate establishes data truth, institutional authority, population fairness or remedy. The generalizable design obligation is to evaluate these distinct obligations jointly before export and to preserve a reason for rejection; the experiment tests this obligation for the specified contract rather than proving universal novelty.

4.5 Candidate search and MAS

Candidates in the reported experiment reuse fixed scores and vary policy only. We distinguish the full 80-point (K, γ) grid from its 16-point fixed-K subset. Each candidate has an ID defined by dataset, capacity and threshold. The runner supports an explicit policy grid and stores all candidates and the selection report.

Pareto dominance is exact over maximize PosRate, minimize absolute gap and minimize abstention. Equivalent candidates remain on the frontier. Fairness-first orders gap, PosRate, abstention; utility-first orders PosRate, gap, abstention. All reported tolerances are zero; the candidate ID is the final tie-breaker. Each analysis records these conventions alongside its candidate domain and selected policy. Repeated selection under an unchanged policy must be invariant to input row order. Changing criterion order tests policy sensitivity, not determinism by itself.

5 DESIGN PROPERTIES AND THEIR LIMITS

G1: Rule-aligned audit. Metrics and constraint checks use the decision states emitted by the specified rule. A training-score metric cannot substitute for evaluation of that rule. The analysis identifies the policy and schema associated with each output.

G2: Reproducible policy selection. Given the same scored cases, IDs, candidate domain, objectives, policy and numerical conventions, the implementation applies exact Pareto filtering and deterministic tie-breaking. Tests include permuted cases and candidates. This property concerns computational reproducibility under fixed inputs; new training samples and institutional preferences can change the result.

G3: Explicit failure. An empty feasible set or insufficient evidence closes the schema 2.0 export gate while preserving candidate evidence. Legacy violation records document empirical failures without establishing that such a block occurred. The generated review-required record needs an institutional process to initiate and complete human review.

G4: Evidence integrity. The verifier checks packet hashes, policy content, dataset-manifest bytes, fitted-model bytes, schema fields and decision replay. Hash verification does not require deserializing the model. These checks establish internal consistency within the declared evidence boundary. Source-data truth, external authenticity and meaningful access to recourse require additional safeguards.

All numerical constraints are retrospective sample checks. New cases require monitoring and an appropriate evaluation design. Raising an entropy threshold can increase automation while admitting more errors; the policy analysis measures that trade-off without establishing improved educational outcomes.

6 EXPERIMENTAL SETUP

6.1 Historical data and evaluation scope

We audit fixed scored test cases from three historical runs: ASSISTments interaction correctness ($n = 105107$), OULAD student success ($n = 6519$) and a synthetic binary stress test ($n = 1000$). ASSISTments uses $y = 1$ for a correct response and school-ID parity as a synthetic grouping proxy, not a protected attribute. OULAD uses $y = 1$ for Pass/Distinction and binary gender for auditing. Appendix 13 gives task and split details. These are outcome-prediction testbeds; descending success scores do not prioritize the highest-risk students for intervention.

The historical runner used a seeded shuffled 80/20 split and seed 42. We do not claim a temporal early-warning horizon, student-cluster independence for interaction rows, or prospective treatment effects. We replay the emitted scores without retraining the predictors. Policy search uses labels from the same retrospective audit sample, so selected operating points are exploratory, not an independently validated final policy.

6.2 Policy domains

All datasets use the threshold grid $\{.01, .05, .10, .15, .20, .30, .40, .50, .60, .70, .80, .90, .95, .99, .995, .999\}$ bits. Capacity grids are ASSISTments $\{100, 250, 500, 1000, 2000\}$, OULAD $\{50, 100, 200, 500, 1000\}$ and synthetic $\{20, 50, 100, 200, 300\}$. Each full grid has 80 candidates. Fixed-budget recovery uses only the 16 thresholds at $K = 1000, 500, 100$, respectively. Full-grid profile comparisons are reported separately. We check gap $\le .05$ and automation coverage $\ge .50$ and apply the algorithm in Section 4.

6.3 Exploratory contestation study

The study includes 16 volunteers, eight standardized appeal scenarios and 64 case responses per condition. Profiles include engineering/data science, analytics and audit/compliance. Each participant responds to four cases under A (logs/cards) and four under B (structured per-case evidence, policy provenance and replay narratives). G1 receives odd cases in A and even cases in B, with G2 reversed.

Each case therefore has eight responses per condition. No participant sees the same case in both conditions. Available study records do not document detailed recruitment procedures, compensation or training.

For each case, participants choose a diagnostic label and remediation action, cite evidence, and rate confidence from 1 to 5. Evidence length is counted in Unicode characters, not words or a quality rubric. Time was self-reported for the questionnaire; we do not estimate per-case staff savings. The case materials include hypothetical exceptions and feature changes beyond the fixed-score replay engine. Consequently this study evaluates availability of structured evidence as a bundle, not execution of every scenario by the software or the isolated causal effect of replay.

Consent and ethics reporting. Participation was voluntary, and the forms contain a consent item. The study lead confirmed participant authorization. No formal ethics-committee approval or exemption was obtained. Participant-level responses and free-text evidence are excluded from the public analysis outputs. These records do not establish institutional ethics authorization or permission for public redistribution of individual responses.

Retrospective analysis. We calculate participant-level condition averages from the response CSVs and case scoring keys. We use two-sided signed-rank tests by enumerating all sign assignments to nonzero paired midranks; zero differences are excluded and equal absolute differences receive average ranks. No normal approximation or continuity correction is used. Holm adjustment covers diagnosis, remediation, confidence and evidence length. Paired mean differences have percentile 95% intervals from 10000 participant-bootstrap replicates (seed 42). This analysis was specified after inspecting historical results and is not preregistered. Diagnosis and remediation correctness coincide on all 128 case responses and are not independent replications. Software versions and exact counts accompany the analysis scripts. The eight scoring keys prescribe eight distinct diagnostic categories and eight distinct remediation actions, one pair per scenario. This one-to-one task construction limits construct separation; it does not force respondents to choose both answers consistently. No independently blinded rubric validation is claimed.

6.4 Retrospective selection/test partition

Before computing split-specific results, we recorded a protocol and partitioned each historical scored sample into validation (floor half) and test (remainder), sorting persistent case IDs by a salted SHA-256 digest. The split uses neither outcomes nor audit groups. At the fixed absolute capacities, we select one threshold from the 16-point grid using validation labels only, feasible-Pareto filtering, the fairness-first profile and zero tolerances. We save all partitions and selected policies with hashes before computing test metrics. If validation has no feasible candidate, we do not substitute a test-selected policy.

The same capacity is applied to each half; Top-K is batch-dependent, so this is not an individual decision invariant to population size. We report all test results without retuning. Wilson intervals describe conditional PosRate and automation coverage. Conservative gap intervals combine two 97.5% Wilson TPR intervals

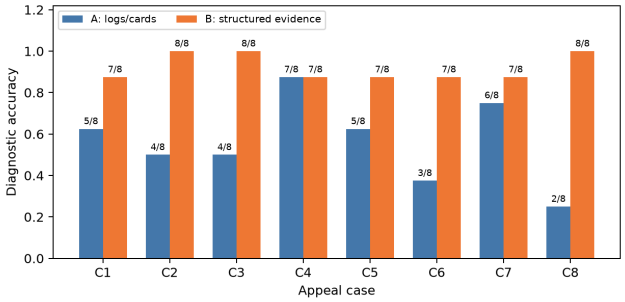


Figure 1: Diagnostic accuracy by case and condition. Labels give exact correct/total counts; all bars have eight responses.

by Bonferroni. These are row-based diagnostics, not cluster-robust guarantees. The historical data and grid were previously inspected: this is a retrospective policy-transfer check on fixed scores, not a pristine prospective holdout, a new predictor fit or a correction for student-level leakage in the original split.

6.5 Synthetic fault contracts

We compare three explicitly limited contracts: passive logs/cards retain information without automated rejection checks; a replay-only checker compares reconstructed decision states with emitted states; the full checker additionally verifies schema, byte hashes, the expected case roster and empirical constraints before authorization. Seven synthetic fixtures cover an unchanged batch, an un-rehashed score edit preserving decisions, an omitted unselected case, changed model bytes, incompatible entropy units, missing policy content and a fairness-violating but replayable policy. This is a functional comparison of specified checks, not an ablation of participant performance or a benchmark against all governance toolkits.

7 RESULTS

We report human task performance, policy analyses on preserved scores, and synthetic tests of the export contract. Numerical tables are generated from case packets or study responses. The policy analyses reuse the predictors' emitted scores.

7.1 Structured evidence in the exploratory study

Table 5 reports condition averages, paired effects and multiplicity-adjusted statistical tests. Diagnostic and remediation accuracy increase from 0.5625 to 0.921875, a difference of 0.359375. Confidence increases from 3.078125 to 4.234375. The paired analysis uses the two-sided sign-enumeration procedure and Holm correction specified in Section 6.3. Longer evidence statements are descriptive and do not establish better reasoning. The experiment supports a benefit of the complete evidence bundle on these standardized tasks, with the limitations in Section 6.3.

Table 5: Exploratory N=16 study; 64 case responses per condition. Participant means and paired differences with 95% participant-bootstrap intervals. p_H is the two-sided signed-rank sign-enumeration p-value adjusted by Holm over four endpoints; diagnosis/remediation correctness coincides in all 128 responses.

Outcome	A	B	Difference [95% CI]	p_H
Diagnosis	0.562	0.922	0.359 [0.203, 0.516]	0.00293
Remediation	0.562	0.922	0.359 [0.203, 0.516]	0.00293
Confidence	3.078	4.234	1.156 [0.922, 1.375]	0.00012
Evidence (characters)	69.812	91.656	21.844 [20.047, 23.734]	0.00012

Table 6: Diagnostic counts by case and condition. Every cell has eight responses; case materials are standardized research scenarios.

Case	A	B
C1	5/8	7/8
C2	4/8	8/8
C3	4/8	8/8
C4	7/8	7/8
C5	5/8	7/8
C6	3/8	7/8
C7	6/8	7/8
C8	2/8	8/8

7.2 Initial decision states and capacity

Table 7 reports the initial recipe outcomes and Table 8 gives exact counts. In the synthetic example $K = 100$ but only four cases are eligible and selected; coverage 0.004 is therefore consistent with the capacity ceiling. OULAD similarly selects 69 of a possible 500. These outputs use the legacy runner, which records violations while retaining candidate decisions; it does not establish an export block or delivered human review.

7.3 Fixed-budget recovery and variable-budget profiles

Table 9 separates full-grid and fixed-budget counts. Table 10 reports achieved metrics of the fixed-budget MAS choices. Raising γ recovers sample feasibility without retraining, demonstrating a policy-level route to resolve the particular violation. This does not exclude simultaneous model or measurement problems. Table 11 reports separate variable-budget comparisons with the complete profile definitions in Section 4. Full precision, group counts, exact grids and candidate IDs are in the structured computational report.

7.4 Taxonomy and uncertainty

Table 12 fixes scores, K and γ while changing abstention semantics. The selected-set positive outcome rate stays fixed in this example, but automation coverage changes because non-selection can be an automated rejection or a deferral. Treating this as a policy correction requires institutional justification; the numbers alone cannot establish which responsibility assignment is appropriate.

Table 13 reports calibration and entropy/error diagnostics using entropy in bits and average ranks for ties. Here ρ correlates entropy with $|y - s|$, the probabilistic absolute error, rather than the binary classification-error indicator. OULAD has lower entropy/absolute-error monotonicity than ASSISTments; the synthetic predictor is poorly calibrated. Temperature scaling can worsen ECE, so we do not claim calibration robustness from preserved rank correlation. The report includes conditional row-bootstrap intervals and group-specific error counts and Wilson intervals across thresholds. Historical interaction packets lack student cluster IDs, so the ASSISTments intervals do not establish cluster-robust uncertainty or field reliability.

7.5 Artifact checks and appeal scope

The schema 2.0 verifier computes integrity and decision-reconstruction checks, with failures tested after packet tampering or removal of policy content. Fitted-model hashes and configuration hashes identify separate objects. The legacy packets contain a configuration digest and a four-point scorecard populated by constant booleans; these fields establish neither fitted-model identity nor independent artifact validation.

The appeal pack (Table 14) is a set of study materials. Medical exceptions, changing income, or removing a feature require additional policy inputs or re-scoring and are outside the fixed-score replay contract. Evidence strings in these materials are not execution logs proving that the released engine implements those operations.

7.6 Frozen threshold evaluation and failure contracts

Table 15 reports the retrospective partition. ASSISTments and OULAD select $\gamma = 0.999$ bits on validation. Both chosen policies satisfy the empirical constraints in the test half. The synthetic validation half has zero feasible candidates, so no synthetic test policy is evaluated. This failure is retained; no threshold or capacity is adjusted after seeing it.

Conditional 95% selected-set positive outcome-rate intervals in test are [0.9742, 0.9901] for ASSISTments and [0.6667, 0.7461] for OULAD. Conservative absolute-gap intervals are [0, 0.0090] and [0, 0.0974], respectively. Thus OULAD's point gap satisfies the 0.05 cap, but its interval does not establish a population gap below that cap. Exact counts, partition hashes, policies, group denominators and intervals are released. These diagnostics do not establish prospective deployment validity.

Table 7: PosRate is the selected-set positive outcome rate (legacy Utility@K), not intervention benefit. Historical recipe outcomes, recomputed from emitted states. Entropy thresholds are in bits; VIOL indicates a retrospective ECL violation, not a delivered human review.

Dataset	K	γ	PosRate	Coverage	Abstain	Gap	ECL
ASSISTments	1000	0.60	0.981	0.758560	0.241440	0.00497	OK
OULAD	500	0.60	0.174	0.010584	0.989416	0.00192	VIOL
Synthetic	100	0.65	1.000	0.004000	0.996000	0.00895	VIOL

Table 8: Exact initial state counts. K is a capacity ceiling, not the realized number selected.

Dataset	n	Eligible	Select	Reject	Abstain
ASSISTments	105107	79730	1000	78730	25377
OULAD	6519	69	69	0	6450
Synthetic	1000	4	4	0	996

Table 9: Candidate counts under gap ≤ 0.05 and coverage ≥ 0.50. Full grids vary K and γ; fixed-budget analyses use only the 16 thresholds at the recipe budget.

Dataset	Full	Feas.	Fixed	Feas.	Pareto
ASSISTments	80	55	16	11	1
OULAD	80	15	16	3	1
Synthetic	80	12	16	3	1

In the seven fault contracts, the unchanged fixture passes replay, verification and the gate. Four altered fixtures preserve replay agreement (score edit, omitted case, changed model bytes and incompatible units), but fail evidence verification. Missing policy content fails both checks. A fairness-violating fixture passes replay and evidence verification but fails the gate. The omitted-case fixture shows why per-packet hashes require a complete-roster check: hashes of retained packets can remain valid after a case is removed. The result demonstrates the difference between consistency, completeness and authorization under the specified contract, not external authenticity or institutional remedy.

Under the predicates in Section 4.4, the unchanged fixture has $(R, I, A) = (1, 1, 1)$; the four evidence faults have $(1, 0, 0)$; missing policy has $(0, 0, 0)$; and the constraint violation has $(1, 1, 0)$. The patterns $(1, 0, 0)$ and $(1, 1, 0)$ refute sufficiency of replay for full verification and of full verification for export permission, respectively; they do not demonstrate logical independence. The expected/observed checks in Table 16 make the synthetic comparison explicit. Passive logs/cards have no automatic rejection contract here and are not assigned a measured detection score. Blocking an export is the observed software consequence; prevented institutional harm and delivered human review were not evaluated.

8 DISCUSSION AND LIMITATIONS

8.1 Institutional value and responsibility

An audit trail makes choices inspectable, but does not make the choices legitimate by itself. Constraints are set by actors with authority and incentives; versioning can expose changes but cannot prevent capture or selective compliance [15, 20]. Institutions still need accessible appeal channels, a mandate for policy changes, appropriate staffing and a way to reverse or remedy decisions. A deployment should record who changed a policy, why, who may contest it and how resulting cases are handled.

Automation coverage measures automated outcomes, including rejection. It must not be equated with access to support or students helped. In screening mode the coverage floor can conflict structurally with limited K; increasing automation by relabeling cases changes the division of responsibility between software and staff. Our experiments expose that choice rather than endorse one taxonomy universally.

8.2 What the evidence establishes

The strongest evidence is computational: rules can be made explicit, decisions can be replayed on fixed scores, infeasibility can be surfaced, and exports can be gated on the declared contract. Sample-level constraints are not population guarantees. The historical full-sample policy comparisons remain exploratory. The retrospective selection/test partition prevents direct use of test labels for threshold selection in that execution, but uses previously inspected scores and does not replace prospective, temporal or student-disjoint evaluation. One historical model and split per dataset do not establish multi-seed predictive stability, comparative state of the art or educational treatment benefit.

The N=16 study is exploratory, uses author-defined standardized scenarios and mixed technical profiles, and has incomplete recruitment and training documentation. Volunteers and consent do not substitute for a formal ethics review, which was not obtained. No causal separation among provenance, structured information and replay is possible in its two conditions. The one-to-one diagnostic/remediation keys also limit outcome independence. A future component ablation must hold information content comparable, specify scoring independently and document task order and learning effects before collecting new responses. Diagnosis and remediation correctness are perfectly concordant in these responses, so the study provides fewer independent outcome dimensions than its labels might suggest. A prospective study with independent

Table 10: PosRate is the selected-set positive outcome rate (legacy Utility@K), not intervention benefit. MAS selections from the fixed-K feasible Pareto set. All tie tolerances are zero; persistent candidate ID is the final tie-breaker.

Dataset	K	γ	Selected	PosRate	Gap	Coverage	Abstain
ASSISTments	1000	0.999	1000	0.981	0.00497	0.999448	0.000552
OULAD	500	0.999	500	0.732	0.01345	0.906427	0.093573
Synthetic	100	0.999	100	1.000	0.01697	0.933000	0.067000

Table 11: PosRate is the selected-set positive outcome rate (legacy Utility@K), not intervention benefit. Variable-K profiles over the full 80-point grids; these are distinct from fixed-budget recovery. Fairness-first orders gap, utility, abstention; utility-first orders utility, gap, abstention.

Dataset	Profile	K	γ	PosRate	Gap	Coverage
ASSISTments	Fairness-first	100	0.999	0.980	0.00040	0.999448
ASSISTments	Utility-first	250	0.999	0.992	0.00206	0.999448
OULAD	Fairness-first	50	0.999	0.800	0.00685	0.906427
OULAD	Utility-first	100	0.999	0.830	0.00754	0.906427
Synthetic	Fairness-first	20	0.999	1.000	0.00020	0.933000
Synthetic	Utility-first	20	0.999	1.000	0.00020	0.933000

Table 12: PosRate is the selected-set positive outcome rate (legacy Utility@K), not intervention benefit. ASSISTments taxonomy replay with fixed scores, $K = 1000$ and $\gamma = 0.60$ bits. Changes in automation coverage represent changes in decision responsibility, not demonstrated improvement in student outcomes.

Policy	PosRate	Coverage	Abstain	Gap
Global abstention	0.981	0.758560	0.241440	0.00497
Top-K-only abstention	0.981	1.000000	0.000000	0.00497
Defer remainder	0.981	0.009514	0.990486	0.00497

Table 13: Recomputed entropy validation. ECE uses 10 equal-width probability bins including score 1; Spearman $\rho = \mathrm{corr}_{\mathrm{rank}}(u, |y-s|)$ uses average ranks for ties; the error is probabilistic absolute error, not binary misclassification. Both baseline and temperature stress use entropy in bits.

Dataset	ECE	ρ	ECE ($T = 2$)	ρ ($T = 2$)
ASSISTments	0.0271	0.9789	0.0905	0.9818
OULAD	0.0128	0.4166	0.0703	0.4166
Synthetic	0.2914	0.6774	0.2398	0.6774

cases, frontline stakeholders and documented institutional review is needed before making field claims.

8.3 Uncertainty and data limitations

Predictive entropy is a scalar confidence proxy, not a decomposition of epistemic and aleatoric uncertainty. Calibration and entropy/error correlation do not alone validate the safety of deferral. We therefore provide group-level selective-error summaries and explicitly conditional intervals. ASSISTments uses school-ID parity only to stress the constraint machinery; it does not measure protected-group disparities. OULAD uses binary gender and success labels, leaving wider intersectional and measurement questions unresolved [3].

8.4 Privacy, integrity and operational boundaries

Packets must contain the inputs required for the supported rule, with purpose-appropriate access and retention controls. Publishing medical information or raw feature histories is not required by our minimum contract. An institution can keep restricted source evidence behind resolvable references, but missing evidence must be reported rather than treated as replay success. Hashes establish consistency of bytes, not external authenticity or truth. The runner's review-required file is not a connected case-management process.

We make no automatic legal-compliance claim. A particular deployment requires jurisdiction-specific institutional review. Instrumented per-case audit cost, staffing capacity, actual access to appeal, independent prospective evaluation and model re-scoring under corrected inputs remain future work. Changing a feature or target creates a new inference or modeling task and must not be described as a policy-only replay.

Table 14: Illustrative N=16 study materials, not execution logs. Full Portuguese narratives and A/B evidence strings remain in the eight local case JSONs. The scope column distinguishes fixed-score replay from operations that the engine does not implement.

Case	Diagnostic category	Evidence illustrated in the instrument	Engine scope
C1	Stale input data	Income snapshot and update lineage	Corrected inputs require new inference.
C2	Policy change	Policy versions and effective dates	Fixed-score replay only for supported policy fields.
C3	Normative exception	Medical flag and force-inclusion rule	Exception rule and medical inputs are not implemented.
C4	Capacity constraint	School capacity and revised availability	Global K replay supported; per-school capacity is outside the engine.
C5	Fairness constraint	Group gap and policy cap	Retrospective candidate check; no per-person fairness reranking.
C6	Proxy risk	ZIP-code correlation and feature removal	Proxy audit/removal requires additional analysis or fitting.
C7	Abstention error	Uncertainty value and threshold	Supported score-based abstention reconstruction.
C8	Version mismatch	Declared and executed policy hashes	Content/hash consistency check; no CI/CD system integration.

Table 15: PosRate is the selected-set positive outcome rate (legacy Utility@K), not intervention benefit. Retrospective selection/test partition of previously inspected fixed scores. The threshold is chosen using validation labels only; the same absolute capacity is used in each half. Test results never retune the policy. This is not a pristine prospective holdout or a refit of the predictor.

Dataset	Split	n	K	γ (bits)	PosRate	Coverage	Gap / status
ASSISTments	validation	52553	1000	0.999	0.983	1.000	0.0000 / selected
ASSISTments	test	52554	1000	0.999	0.984	0.999	0.0026 / feasible
OULAD	validation	3259	500	0.999	0.682	0.912	0.0140 / selected
OULAD	test	3260	500	0.999	0.708	0.901	0.0285 / feasible
Synthetic	validation	500	100	–	–	–	no feasible candidate
Synthetic	test	500	100	–	–	–	not evaluated

Table 16: Synthetic fault contracts: expected/observed (E/O), where 1 means agreement or permission and 0 means rejection. Replay compares available decision states; integrity includes replay, hashes, schema and the expected roster. Export is technical permission under declared checks, not institutional legitimacy. Passive logs/cards have no automated rejection contract in this comparison.

Fixture	Replay E/O	Integrity E/O	Export E/O	Reason for export result
Unchanged control	1/1	1/1	1/1	No block
Score edit preserving decisions	1/1	0/0	0/0	Packet hash mismatch
Unselected case omitted	1/1	0/0	0/0	Expected-roster mismatch
Model bytes changed	1/1	0/0	0/0	Model digest mismatch
Incompatible entropy unit	1/1	0/0	0/0	Unit/schema check
Policy content missing	0/0	0/0	0/0	Policy unavailable
Fairness-violating policy	1/1	1/1	0/0	Declared empirical cap

9 CONCLUSION

Compass-ECL-MAS connects explicit decision policies, empirical constraints, deterministic feasible-Pareto selection and fixed-score replay through an executable export contract. Its acceptance conditions distinguish reproduction of supplied decisions from verification of complete evidence and from satisfaction of declared export constraints. Seven synthetic fault fixtures demonstrate why replay agreement alone is insufficient for technical export permission.

The educational testbeds show that policy thresholds, capacity and abstention semantics materially affect sample feasibility and automation coverage. The retrospective selection/test analysis preserves the synthetic case's infeasibility instead of selecting a policy on test data. In the exploratory N=16 study, the structured-evidence condition supports higher accuracy on standardized diagnosis and remediation tasks, while the bundled intervention and concordant correctness endpoints limit attribution to individual components.

These findings support an infrastructure contribution: inspectable policy choices, bounded decision reconstruction and explicit rejection when the export contract fails. Institutions must connect that infrastructure to accessible appeals, authorized policy decisions and effective remedies. Prospective evaluation with stakeholders, documented ethical review, independently specified scoring and an instrumented human workflow are needed to establish its value in practice. Appendix 10 describes the computational artifacts and the access restrictions on participant responses.

Acknowledgments

We thank the volunteers and colleagues who contributed to the exploratory evaluation.

REFERENCES

[1] Kars Alfrink, Ianus Keller, Gerd Kortuem, and Neelke Doorn. 2023. Contestable AI by Design: Towards a Framework. *Minds and Machines* 33 (2023), 613–639. https://doi.org/10.1007/s11023-022-09611-z

[2] Anastasios N. Angelopoulos and Stephen Bates. 2021. A Gentle Introduction to Conformal Prediction and Distribution-Free Uncertainty Quantification. *arXiv preprint arXiv:2107.07511* (2021).

[3] Ryan S. Baker and Aaron Hawn. 2022. Algorithmic Bias in Education. *International Journal of Artificial Intelligence in Education* 32, 4 (2022), 1052–1092. https://doi.org/10.1007/s40593-021-00285-9

[4] C. K. Chow. 1970. On optimum recognition error and reject tradeoff. *IEEE Transactions on Information Theory* 16, 1 (1970), 41–46.

[5] Danielle Keats Citron and Frank Pasquale. 2014. The Scored Society: Due Process for Automated Predictions. *Washington Law Review* 89, 1 (2014), 1–33. https://digitalcommons.law.uw.edu/wlr/vol89/iss1/2/

[6] Virginia Dignum. 2018. Ethics in artificial intelligence: introduction to the special issue. *Ethics and Information Technology* 20, 1 (2018), 1–3. https://doi.org/10.1007/s10676-018-9450-z

[7] District Court of The Hague. 2020. Judgment on the System Risk Indication (SyRI) legislation (ECLI:NL:RBDHA:2020:1878). https://uitspraken.rechtspraak.nl/details?id=ECLI%3ANL%3ARBDHA%3A2020%3A1878 Judgment date: 5 Feb 2020; accessed 2026-01-24.

[8] European Commission, High-Level Expert Group on Artificial Intelligence. 2019. Ethics Guidelines for Trustworthy AI. European Commission Digital Strategy. https://digital-strategy.ec.europa.eu/en/library/ethics-guidelines-trustworthy-ai

[9] Todd Feathers. 2023. False Alarm: How Wisconsin Uses Race and Income to Label Students "High Risk". The Markup. https://themarkup.org/machine-learning/2023/04/27/false-alarm-how-wisconsin-uses-race-and-income-to-label-students-high-risk Accessed 2026-01-24.

[10] Mingyu Feng, Neil T. Heffernan, and Kenneth R. Koedinger. 2009. Addressing the assessment challenge in an Intelligent Tutoring System that tutors as it assesses. *User Modeling and User-Adapted Interaction* 19, 3 (2009), 243–266. https://doi.org/10.1007/s11257-009-9063-7

[11] Yonatan Geifman and Ran El-Yaniv. 2019. SelectiveNet: A deep neural network with an integrated reject option. In *Proceedings of the 36th International Conference on Machine Learning (ICML)*.

[12] Jakub Kuzilek, Martin Hlosta, and Zdenek Zdrahal. 2017. Open University Learning Analytics dataset. *Scientific Data* 4 (2017), 170171. https://doi.org/10.1038/sdata.2017.171

[13] Ben Laurie, Emilia Messeri, and Rob Stradling. 2021. Certificate Transparency Version 2.0. RFC 9162, Internet Engineering Task Force. https://doi.org/10.17487/RFC9162

[14] Timothy Lebo, Satya Sahoo, and Deborah McGuinness. 2013. PROV-O: The PROV Ontology. W3C Recommendation, 30 April 2013. https://www.w3.org/TR/2013/REC-prov-o-20130430/

[15] Jacob Metcalf, Emanuel Moss, Elizabeth Anne Watkins, Ranjit Singh, and Madeleine Clare Elish. 2021. Algorithmic Impact Assessments and Accountability: The Co-Construction of Impacts. In *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency (FAccT '21)*. Association for Computing Machinery, 735–746. https://doi.org/10.1145/3442188.3445935

[16] Brent Mittelstadt. 2019. Principles alone cannot guarantee ethical AI. *Nature Machine Intelligence* 1, 11 (2019), 501–507. https://doi.org/10.1038/s42256-019-0114-4

[17] OECD. 2019. Recommendation of the Council on Artificial Intelligence. OECD Legal Instruments. https://legalinstruments.oecd.org/en/instruments/oecd-legal-0449

[18] Alexandru A. Ormenisan, Moritz Meister, Fabio Buso, Robin Andersson, Seif Haridi, and Jim Dowling. 2020. Time Travel and Provenance for Machine Learning Pipelines. In *2020 USENIX Conference on Operational Machine Learning (OpML 20)*. USENIX Association. https://www.usenix.org/conference/opml20/presentation/ormenisan

[19] Pedro Saleiro, Benedict Kuester, Loren Hinkson, Jesse London, Anna Stevens, Ari Anisfeld, Kit T. Rodolfa, and Rayid Ghani. 2018. Aequitas: A Bias and Fairness Audit Toolkit. *arXiv preprint arXiv:1811.05577* (2018). https://arxiv.org/abs/1811.05577

[20] Andrew D. Selbst, Danah Boyd, Sorelle A. Friedler, Suresh Venkatasubramanian, and Janet Vertesi. 2019. Fairness and Abstraction in Sociotechnical Systems. In *Proceedings of the Conference on Fairness, Accountability, and Transparency (FAT* '19)*. Association for Computing Machinery, 59–68. https://doi.org/10.1145/3287560.3287598

[21] Elham Tabassi. 2023. Artificial Intelligence Risk Management Framework (AI RMF 1.0). https://doi.org/10.6028/NIST.AI.100-1

[22] UNESCO. 2021. Recommendation on the Ethics of Artificial Intelligence. UNESDOC Digital Library. https://unesdoc.unesco.org/ark:/48223/pf0000380455

[23] Berk Ustun, Yang Liu, and David Parkes. 2019. Fairness without Harm: Decoupled Classifiers with Preference Guarantees. In *Proceedings of the 36th International Conference on Machine Learning (Proceedings of Machine Learning Research, Vol. 97)*. 6373–6382. https://proceedings.mlr.press/v97/ustun19a.html

[24] Manasi Vartak, Harihar Subramanyam, Wei-En Lee, Srinidhi Viswanathan, Saadiyah Husnoo, Samuel Madden, and Matei Zaharia. 2016. ModelDB: A System for Machine Learning Model Management. In *Proceedings of the Workshop on Human-In-the-Loop Data Analytics*. Association for Computing Machinery. https://doi.org/10.1145/2939502.2939516

[25] Hilde Weerts, Miroslav Dudík, et al. 2023. Fairlearn: Assessing and Improving Fairness of AI Systems. https://doi.org/10.5555/3648699.3648956

[26] Maranke Wieringa. 2020. What to Account for When Accounting for Algorithms: A Systematic Literature Review on Algorithmic Accountability. In *Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency (FAT* '20)*. Association for Computing Machinery, 1–18. https://doi.org/10.1145/3351095.3372833

[27] WIRED. 2020. The UK's Exam Algorithm Was a Disaster. Here's Why It's a Warning for AI. https://www.wired.co.uk/article/uk-exam-algorithm-results Accessed 2026-01-24.

10 CODE AND DATA AVAILABILITY

The reference repository is https://github.com/rtales/compass-contestable-governance-artifacts. The executable computational snapshot accompanying this manuscript is `revisao_artigo/rodada4/pacote_avaliador.zip`, identified by SHA-256 `108d5ff9f550dec882b6691b24ec2b5340c5515ab3fd550a198e32e72e947688`. Its manifest binds source, inputs, protocol, scripts and expected outputs. Commit 227f7fada6f98f44a79a6982dfda26413dcd29ac identifies the

Table 17: Focused mechanism comparison with primary sources. This is not an exhaustive capability audit or a measured benchmark of external tools. Entries describe established mechanisms and delimit our own implementation; silence in a source is not treated as proof of an absent feature.

Source	Established mechanism	Relationship and limit of this work
PROV-O [14]	Entities, activities, agents and derivation relations for provenance interchange.	Our decision-packet contract is domain-specific; no implemented PROV-O interchange is claimed.
ModelDB [24]	Model/pipeline history, metadata and metric tracking.	Tracking/versioning itself is established; our evaluated target is the decision contract on fixed scores.
Ormenisan et al. [18]	Change capture and data versions for pipeline reconstruction/debugging.	Our replay does not reconstruct new inference from corrected features.
RFC 9162 [13]	Signed transparency-log structures and inclusion/consistency proofs for certificates.	Local digests and roster checks are weaker: no signed append-only log or independent authenticity.
Alfrink et al. [1]	Sociotechnical framework for responsiveness to intervention across the lifecycle.	Fixed-score scrutiny is a bounded enabling mechanism, without an implemented institutional appeal and remedy.

published base release, not this snapshot. The embedded paper and present manuscript report the same computational results.

Computational reproduction. After extracting the archive, follow its `README.md`: use Python 3.12, install `requirements.lock`, and run `python reproduce.py --output-dir actual`. The runner verifies the manifest, regenerates four JSON reports and compares them structurally with the expected outputs, excluding Python patch-version metadata. It runs 20 application tests and checks seven export-contract vectors. The authors executed these checks after extraction; external independent execution remains a separate validation step. The lock records the tested environment; other platforms may require compatibility work.

The source packets are ASSISTments, OULAD and synthetic scored runs from 26 January 2026, all seed 42. The snapshot supplies packet inputs, partition IDs, frozen policies, group counts, expected test metrics and fault results. It does not refit those predictors. Synthetic training fixtures used by application tests are separate from the preserved scored runs. Byte hashes and unsigned local manifests establish consistency within the bundle, not external authenticity against coordinated replacement.

Full-sample and uncertainty analyses. The project script `revisao_artigo/scripts/recalcular_evidencias.py` generates full-sample candidate tables, group-risk summaries, study aggregates and the diagnostic figure from the preserved inputs. Outputs are in `paper_assets/revised/`; structured reports record grids, thresholds, capacities, IDs and counts before display rounding. Fixed-budget and full-grid analyses retain distinct candidate domains. Source and environment inventories accompany these analyses.

Participant data and access restrictions. The snapshot excludes participant responses and free text and cannot recompute the human study. That analysis requires locally retained response CSVs and scoring keys. Public materials provide aggregates and methods; generated participants never substitute for restricted responses. Participation and consent records do not establish redistribution rights: individual-data release requires resolution of institutional requirements and sharing permissions. The study analysis and policy-transfer check are retrospective, without prospective preregistration.

The accompanying LaTeX source and computational snapshot have separate inventories. Table 17 compares the implemented mechanisms with established approaches and identifies their limits.

11 IMPLEMENTATION AND EVIDENCE INVENTORY

Preserved scored runs supply metrics, metadata, decision briefs and case packets for each dataset. Policy analyses reuse these scores for sweeps, taxonomy replay and retrospective selection/test evaluation. The schema 2.0 application and synthetic fixtures test evidence verification and export permission. The human study uses separate responses and scenarios; participant data are excluded from the snapshot.

The application source is included under `application/` in the snapshot identified in Appendix 10. Runs persist policy content, a dataset-manifest copy, fitted-model bytes, separate configuration/model hashes, source-file hashes, selection reports, candidate packets, computed evidence checks and an export gate. A Git HEAD alone does not identify uncommitted source changes. Model snapshots are hashed as opaque bytes; verifying integrity does not require loading serialized code.

The gate permits export after verification and feasibility checks. Infeasibility or insufficient evidence retains candidate metrics and produces a review-required record, without authorized export or delivery to a human queue. Legacy decisions and violation records predate the schema 2.0 guarantees.

The evaluation covers fixed-score policies and the verification contract. Predictive-baseline comparisons, multi-seed modeling, field deployment and individually timed appeal costs remain unevaluated.

12 REQUIREMENTS FOR PROSPECTIVE EARLY WARNING

A prospective dropout-prevention task requires a risk-positive label (e.g., Fail/Withdrawn), an intervention horizon, features available before that horizon and aligned ranking. Temporal and student-aware splits, label audits and justified audit groups are also needed. The implemented OULAD success target and descending

ranking evaluate a different task; their PosRate estimates cannot establish early-warning performance.

ASSISTments interaction correctness likewise differs from per-student support need. Aggregation, eligibility, treatment and actual benefit require separate evaluation. A changed target or feature set may require retraining; changed inputs require new inference. These operations exceed fixed-score replay.

13 DATASETS AND IMPLEMENTED TASKS

13.1 ASSISTments

The 2009–2010 Skill Builder release provides student–problem interactions [10]. The implemented target is response correctness (positive=1). School-ID parity is a synthetic group for testing constraints, not a demographic or protected attribute.

The processor reads the CSV with a deterministic encoding fallback, replaces decoding errors and skips malformed lines. Numeric missing values are filled with zero, including skill and response-time fields; the implementation does not drop rows for those missing fields. Correctness is coerced to an integer and clipped to zero/one; missing labels become zero. These choices can affect measurement validity and should be audited before deployment.

Features include bounded attempt and hint counts, log-transformed response and overlap times, practice opportunity, an original-item indicator, skill-ID remainder, and fixed one-hot categories for tutor mode, answer type and task type. School ID itself is excluded from the feature vector, but that does not establish absence of proxies.

The runner applies a seed-42 shuffled row split (80/20); the audited output has 105107 cases. Features can describe the interaction being predicted. This is not a temporally validated early-warning task. The historical packets lack student-cluster IDs, so neither split independence nor cluster-robust intervals are established.

13.2 OULAD

OULAD provides anonymized student/course information [12]. The processor uses the student-information table. Positive means Pass or Distinction; Fail and Withdrawn are negative. Gender F/M defines the binary audit group, with other or missing gender excluded.

Categorical inputs are module, presentation, region, education, deprivation band, age band and disability; numerical inputs are previous attempts and studied credits. Gender and final outcome are excluded from the score features. This is a success-prediction benchmark; it does not use a validated early-intervention time window or estimate treatment benefit. The shuffled 80/20 row split yields 6519 audited cases. Repeated student/course records are not claimed to be separated by student.

13.3 Synthetic

The synthetic audit has 1000 held-out cases and a binary generated group. Its purpose is to exercise capacity, abstention, infeasibility and evidence checks under a controlled label process. It is not evidence of field validity. Recipe inputs, hashes and exact group counts are included in the computational report.

14 ARTIFACT CONTRACT AND REPLAY PROTOCOL

14.1 Minimum fixed-score contract

Schema 2.0 records a persistent case ID, finite score, entropy in bits, clipping convention, decision state, supported replay scope, policy reference/hash, dataset-manifest hash, fitted-model hash and packet hash. Policy content records capacity, score direction, allocation/screening mode, global/Top-K-only abstention, threshold/unit and selection criteria/tolerances. Hashes alone are not substitutes for accessible policy content. Labels and audit groups are evaluation fields, separate from decision inputs.

Historical packets have a smaller unversioned schema. Their entropy is checked against the known bit-based implementation before reanalysis; a legacy configuration digest is not described as a fitted-model digest. Schema 2.0 requirements apply to the reference application outputs, not retroactively to those legacy packets.

14.2 Reconstruction and validation

Replay reads the same fixed scores and case IDs, applies the declared rule and compares every emitted state with the original. The schema 2.0 verifier additionally checks content hashes, schema/unit fields, policy availability, manifest bytes, model-snapshot bytes and equality of the unique packet-ID roster with the expected roster in run metadata. This detects omissions relative to that metadata; coordinated rewriting of unsigned metadata and all hashes is not proof of authenticity. Missing, altered or incompatible evidence causes failure, not a perfect score by default. Tests include score ties, reordered inputs, boundary thresholds, empty feasible sets and corrupted packets.

14.3 Scope of appeal materials

Medical-force-inclusion rules require the applicable rule, its version, the relevant flag and exception precedence. These are absent from the minimum score-based contract; no executed medical-exception replay is claimed. Likewise, income corrections or feature removal require updated inputs and re-scoring, possibly retraining. The study's evidence strings are scenario materials rather than logs proving these operations were executed. Restricted evidence may remain outside public bundles, but its availability and access conditions must be documented.

14.4 Uncertainty analysis

ECE uses ten equal-width probability bins, including probability one in the final bin. Spearman entropy/absolute-error correlation is between u_i and $|y_i - s_i|$ and uses average ranks for ties; it is not correlation with a binary error indicator. Temperature stress uses logits scaled by $T \in \{0.5, 2\}$ and recomputes entropy in bits. Conditional percentile intervals use 200 row-bootstrap replicates, seed 42. Per-group selective risk is the binary classification error among entropy-eligible cases, not precision among Top-K selections; reports give error counts, denominators and descriptive Wilson intervals. Interaction packets do not preserve student cluster IDs, so these row-based intervals do not establish cluster-robust population uncertainty.